MCML charge pump

Virtuoso ADE L and Calculator evidence

Captured 2026-10-22 from actual Virtuoso 6.1.7 windows.

Numerical solver: ngspice 45.2. Models: FreePDK45 nominal VTG.

Source: checked shivam/tb_mcml_charge_pump/schematic Virtuoso netlist.

Cadence PSF waveforms are displayed and evaluated in ViVA/Calculator.

Measured values

Charge delay (10-edge mean): 13.811 ps

Discharge delay (10-edge mean): 15.447 ps

Paper comparison: 7.56% difference from 16.71 ps

Charge VDD power: 22.528 uW; discharge: 16.097 uW

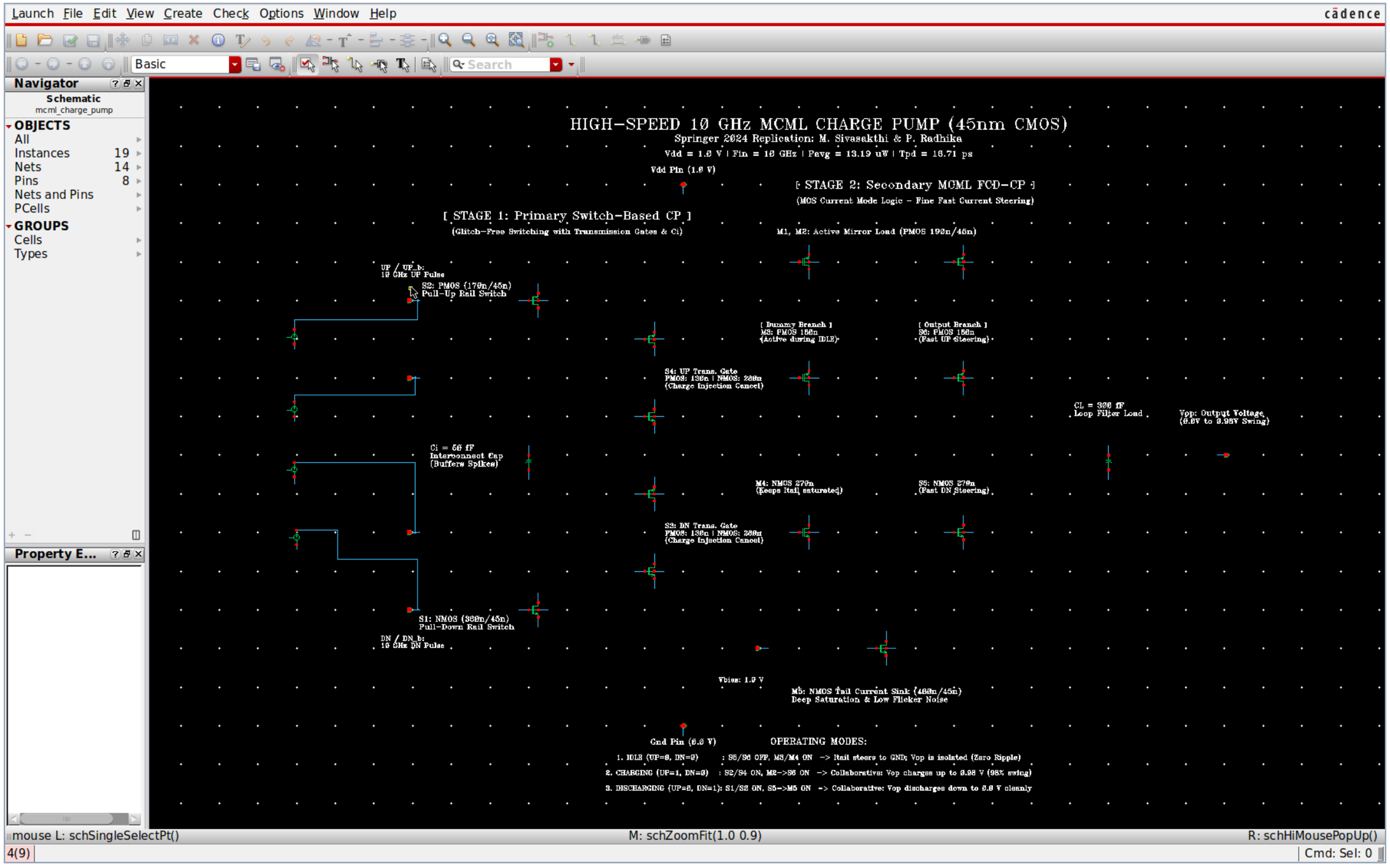
Settled output swing at 80 ns: 993.524 mV

DC mismatch at forced Vop=0.5 V: 115.47%

10% current-compliance criterion: no valid interval containing 0.5 V

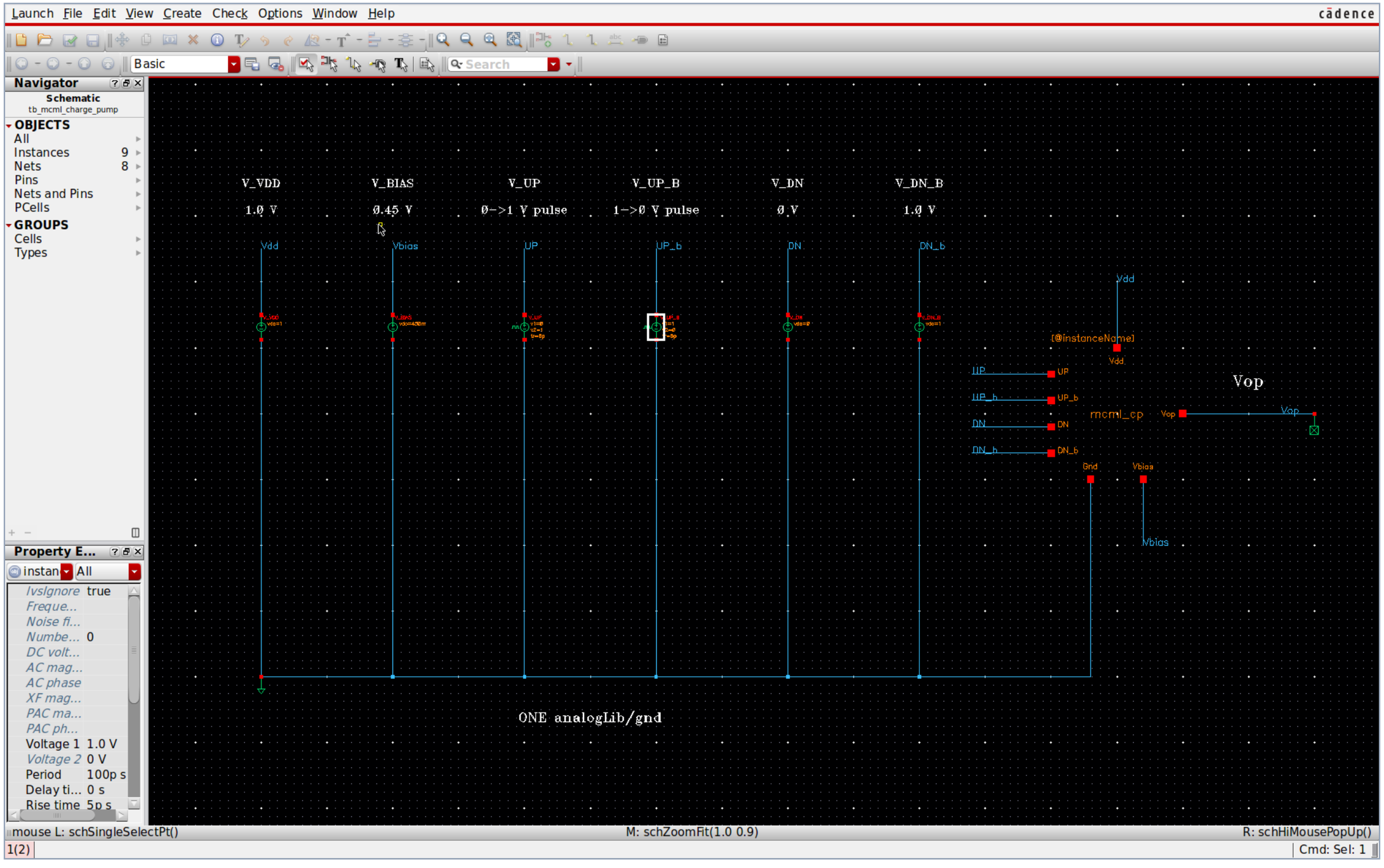
The paper used a 10 pF load and different process models; this project has internal CL=300 fF.

DUT schematic



Actual shivam/mcml_charge_pump/schematic window. The design annotations quote paper targets, not these measured results.
Internal CL=300 fF; the Vbias net drives the M5 tail gate.

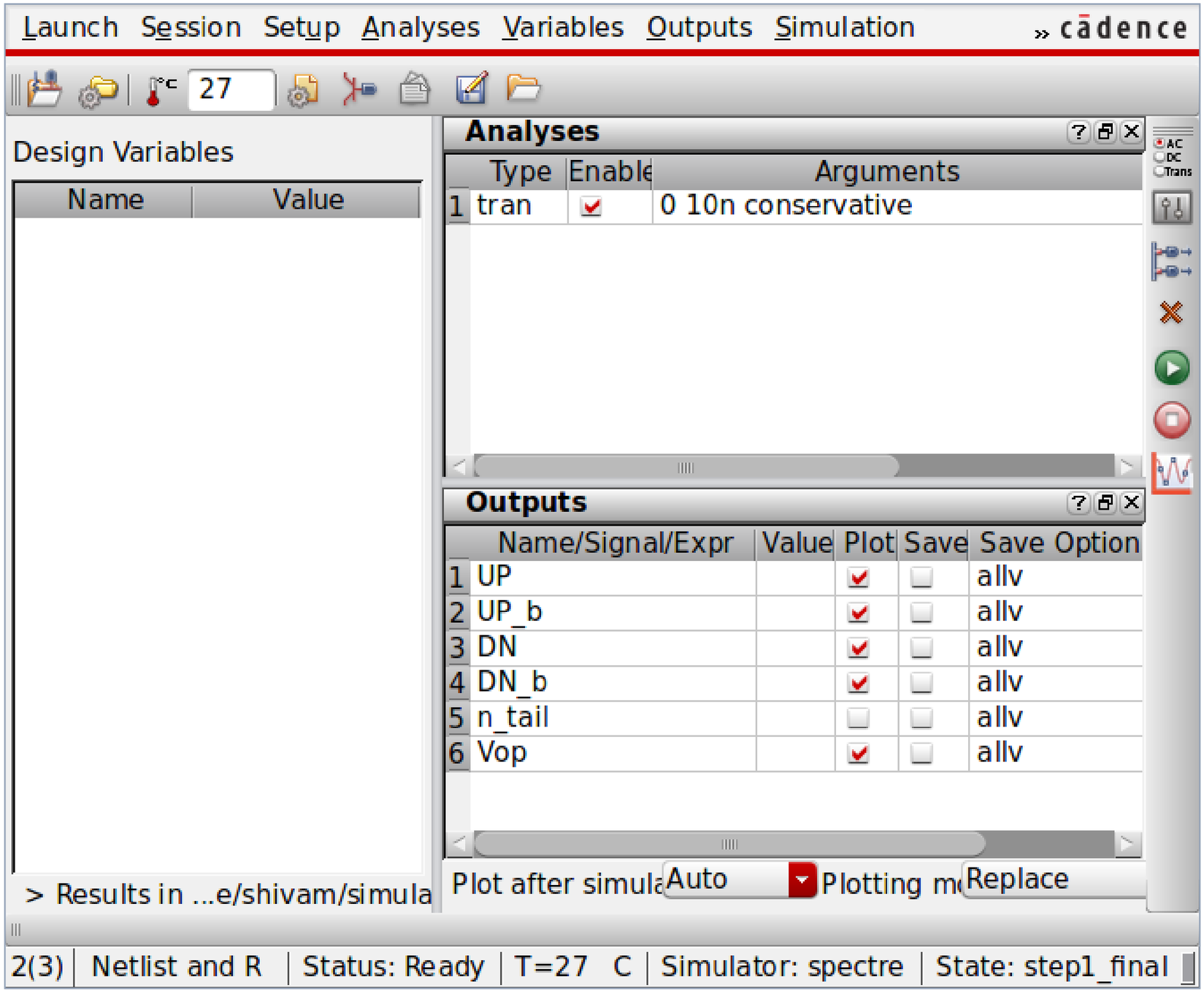
Transient testbench schematic



Actual shivam/tb_mcml_charge_pump/schematic window: VDD=1.0 V, Vbias=0.45 V, UP/UP_b pulses, DN=0 V, DN_b=1.0 V.

Vop is the named output; one analogLib/gnd is connected; no external output capacitor.

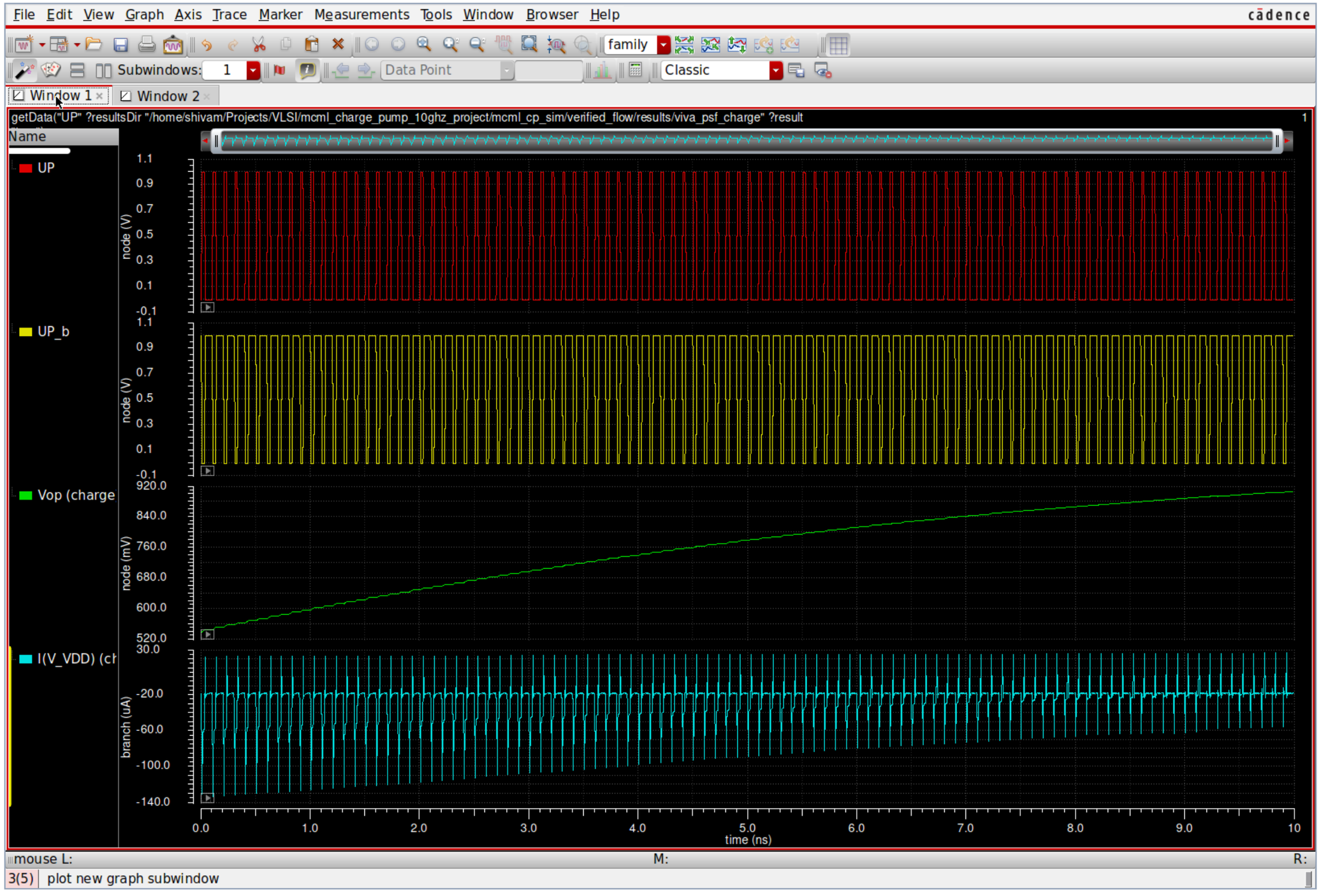
ADE L: step1_final state



The loaded testbench ADE state requests a 10 ns transient at 27 C. Its simulator slot is named spectre by ADE.

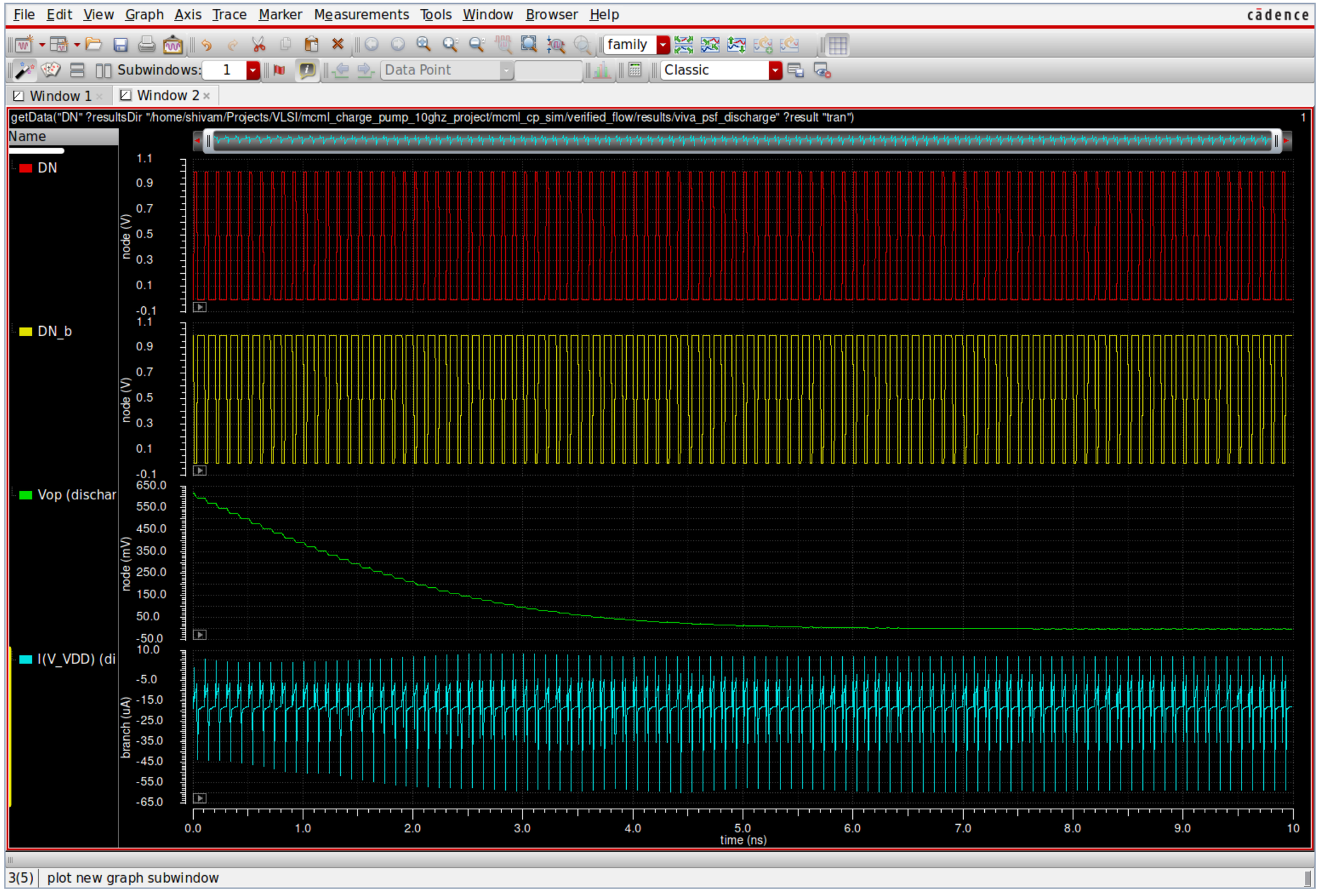
The local adapter executes the numerical simulation with ngspice; the solver is identified in the run logs.

ViVA: charging transient



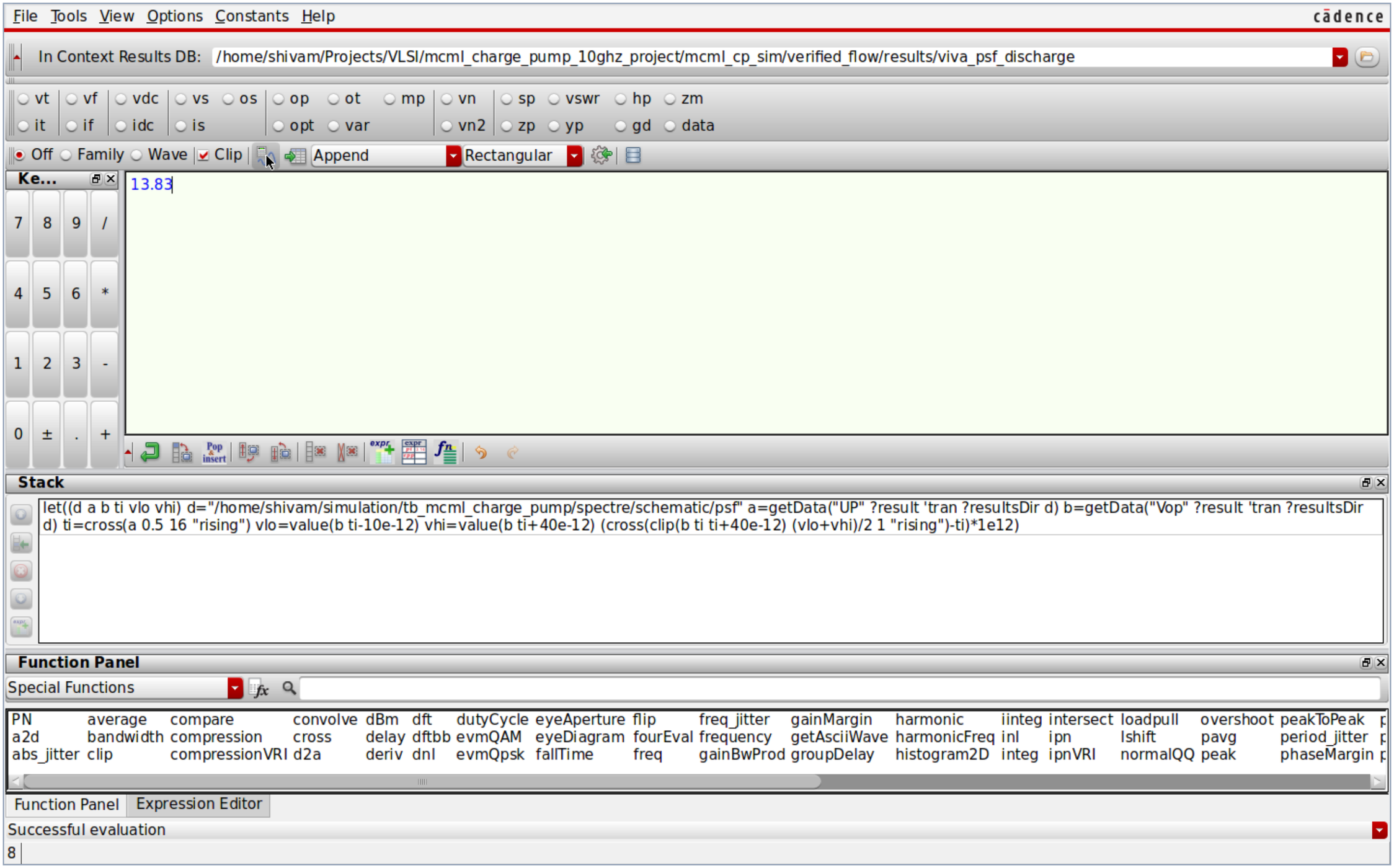
Actual Cadence ViVA display of UP, UP_b, Vop, and VDD source current from the charging PSF data.

ViVA: discharging transient



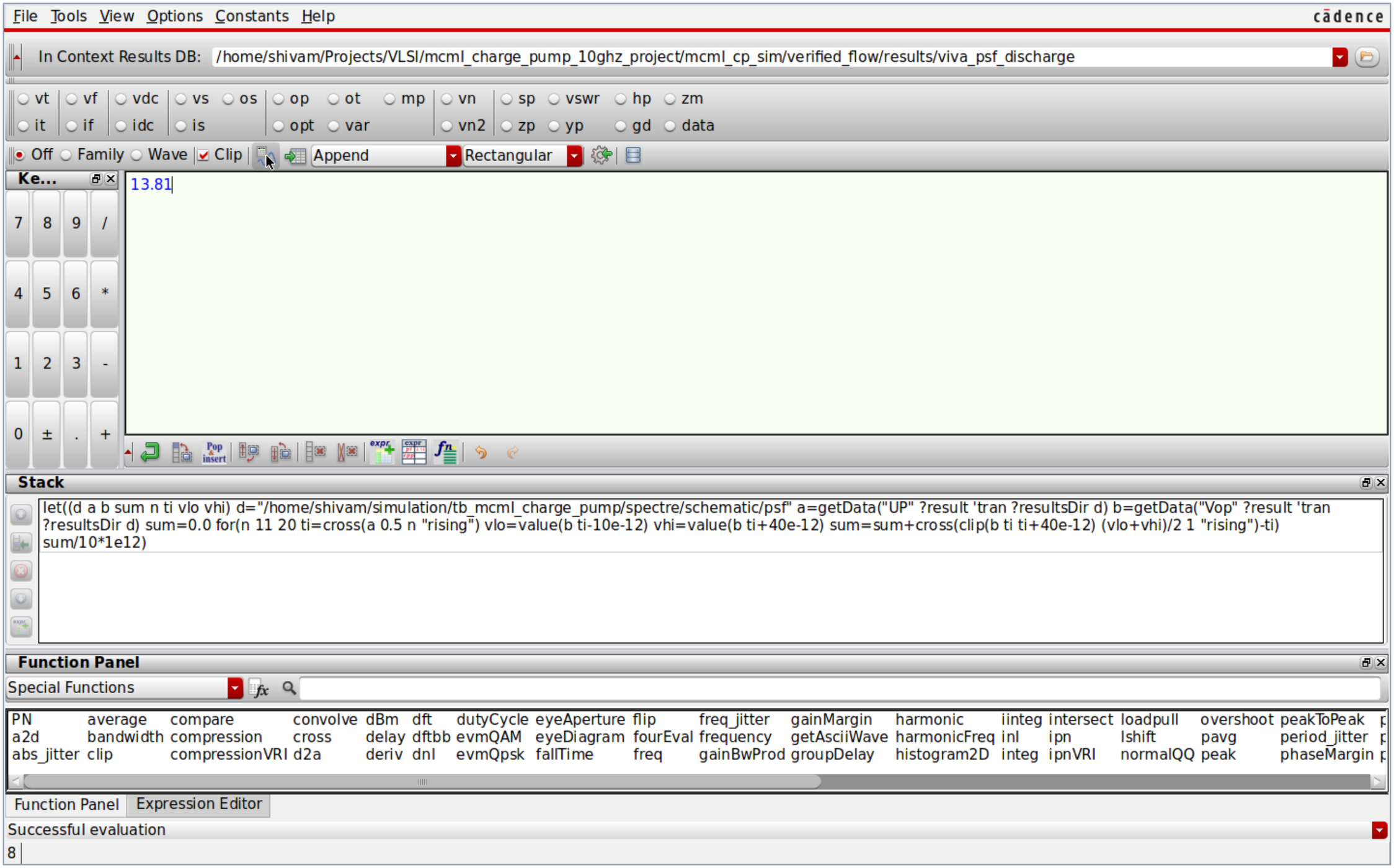
Actual Cadence ViVA display of DN, DN_b, Vop, and VDD source current from the separate discharging PSF data.

Calculator: charge delay, pulse 16



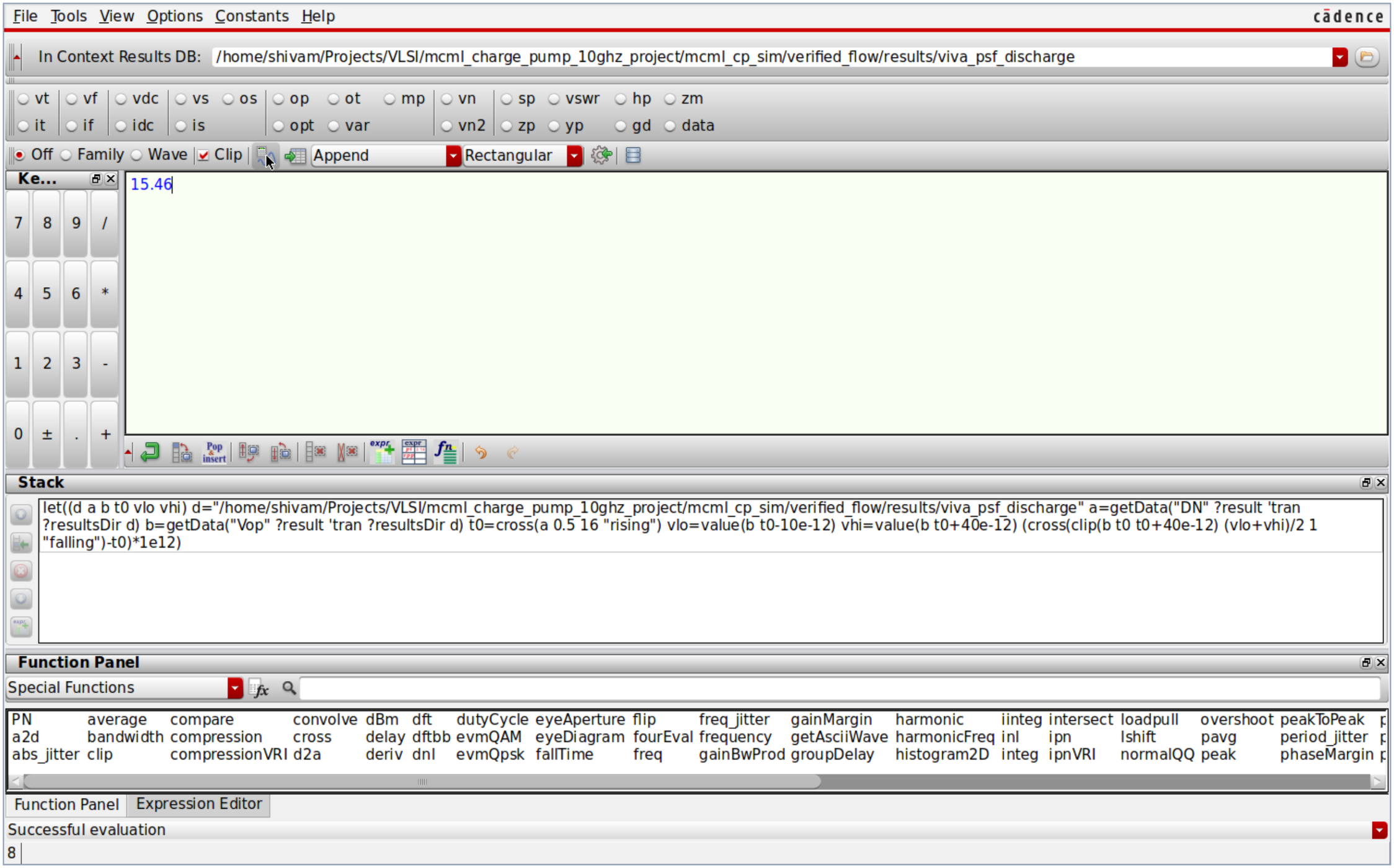
First local midpoint crossing after the UP edge: 13.827 ps (Calculator rounds its display).

Calculator: charge delay, 10-edge mean



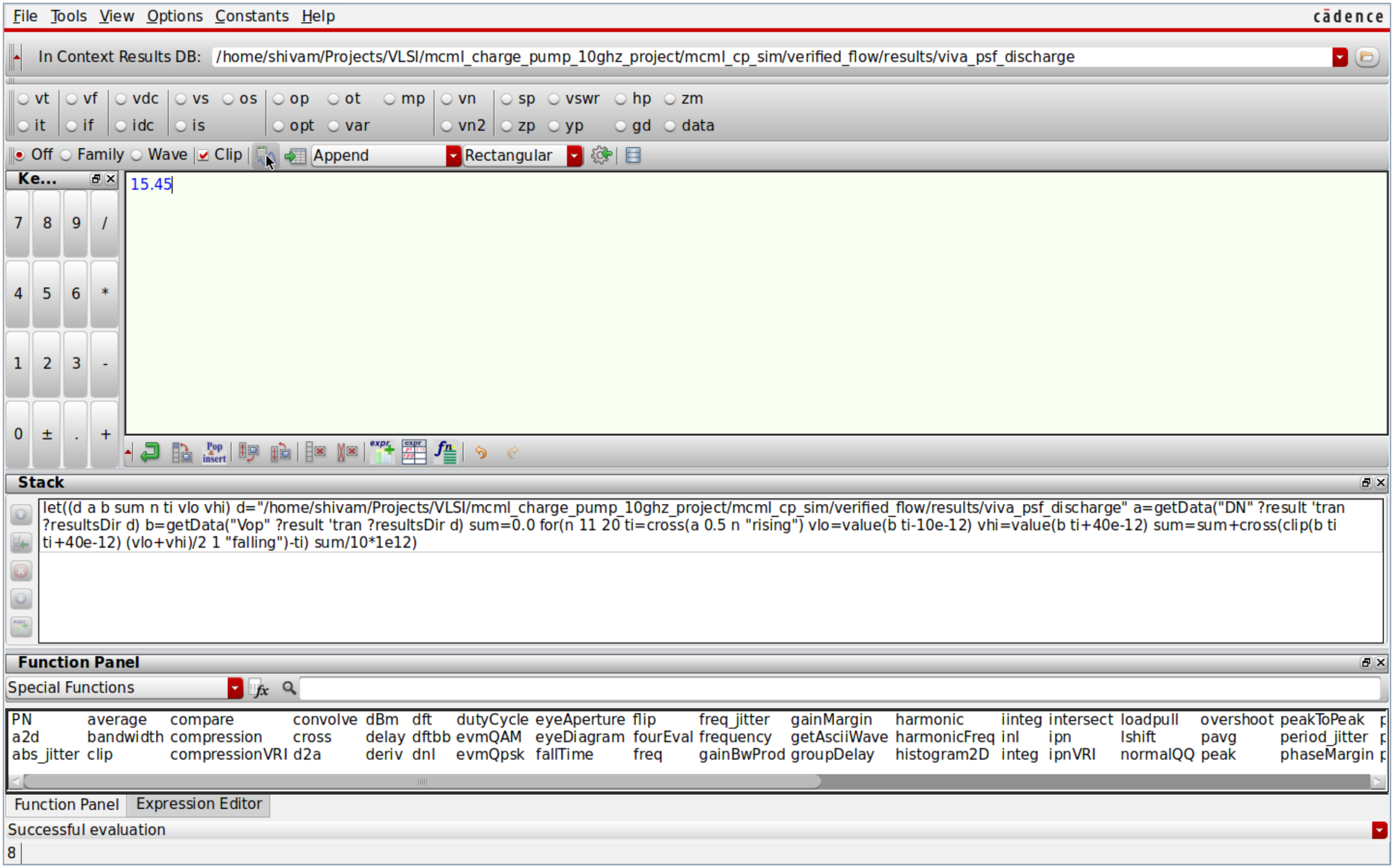
Mean of pulses 11-20: 13.811 ps. The input threshold is 0.5 V.

Calculator: discharge delay, pulse 16



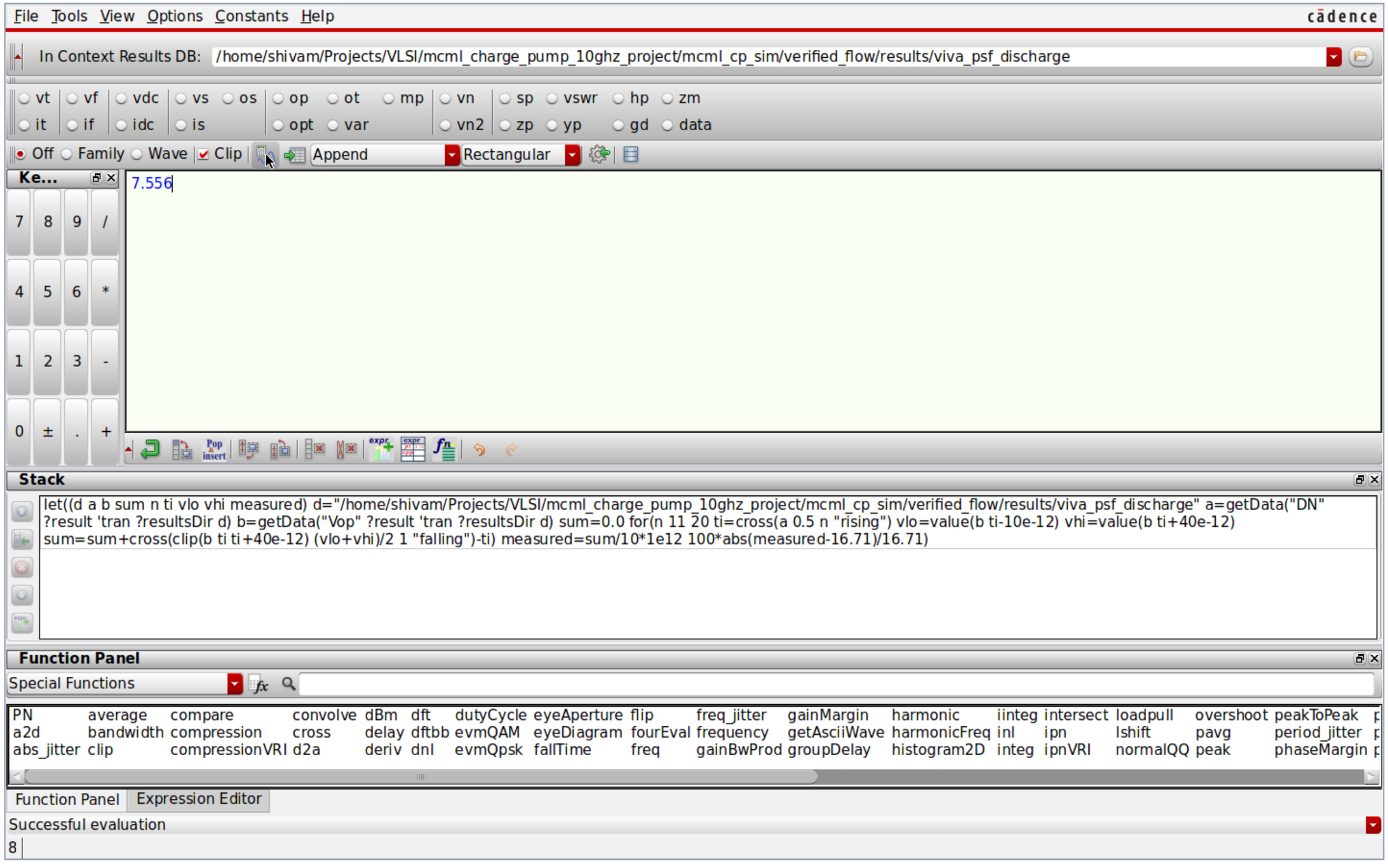
First falling Vop midpoint crossing after the DN edge: 15.460 ps.

Calculator: discharge delay, 10-edge mean



Mean of pulses 11-20: 15.447 ps; this is the larger directional mean.

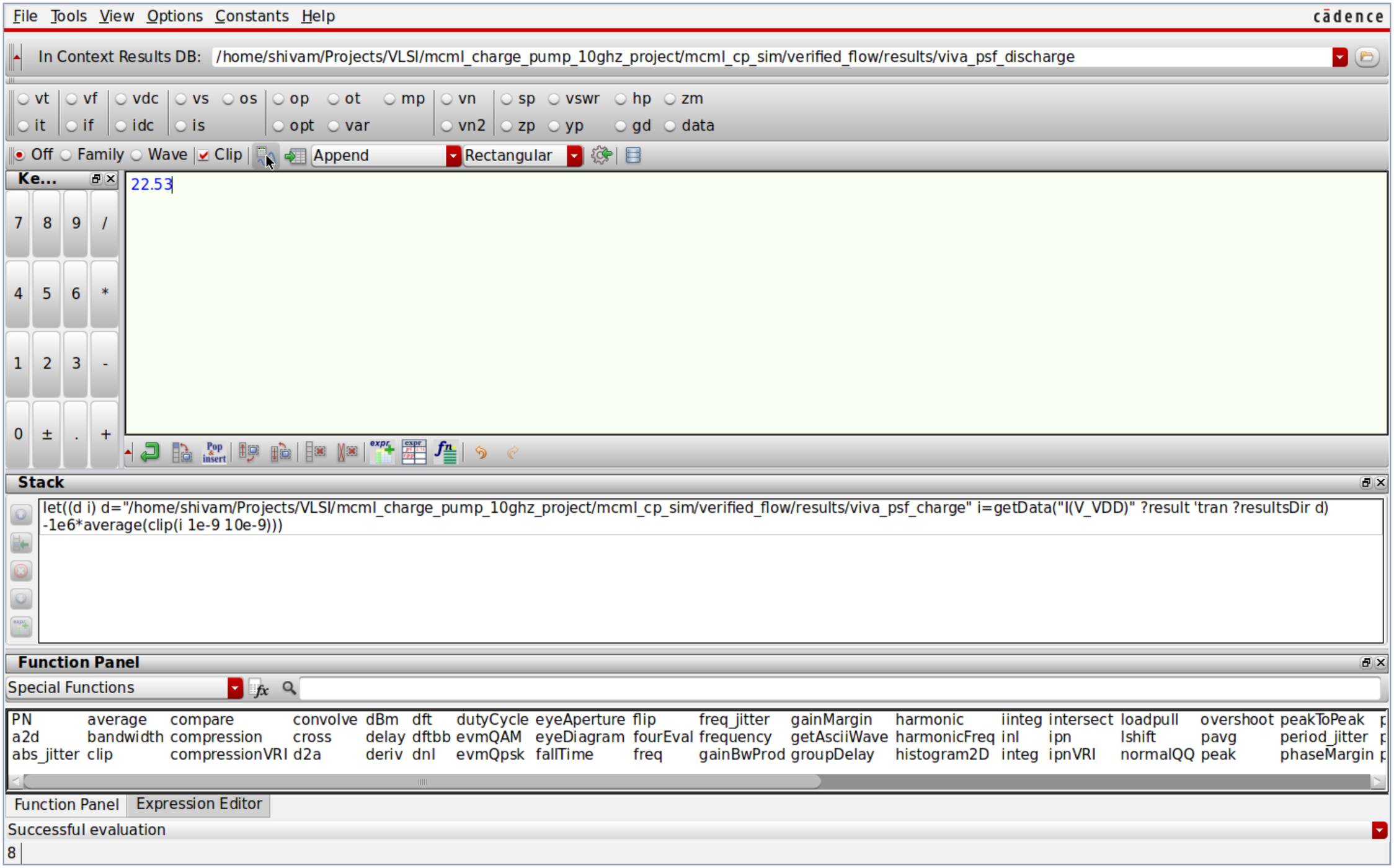
Calculator: difference from paper delay



Absolute difference from the paper's 16.71 ps TT value: 7.56%.

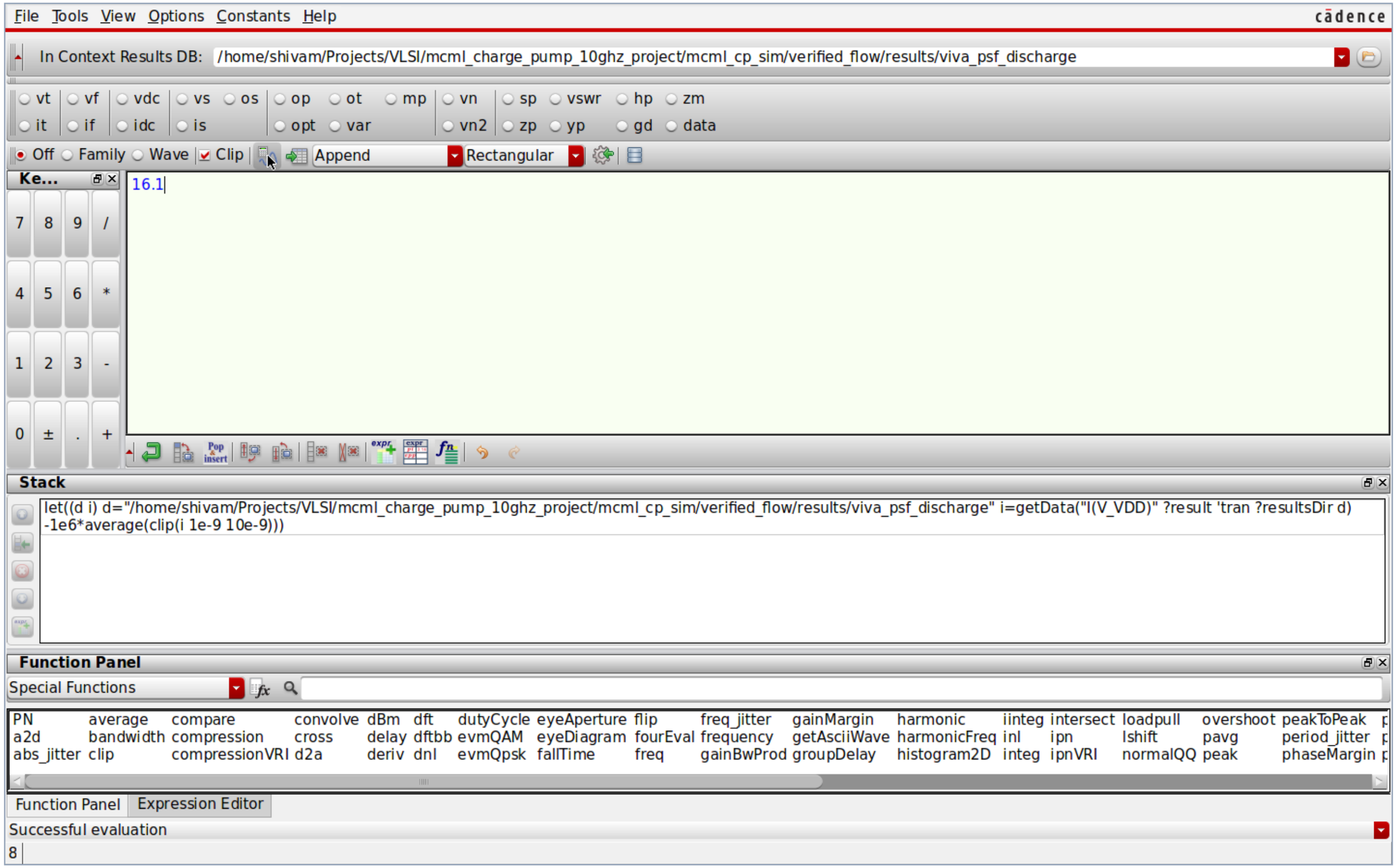
This is numerical proximity under different load and models, not a matched-condition replication.

Calculator: charging VDD power



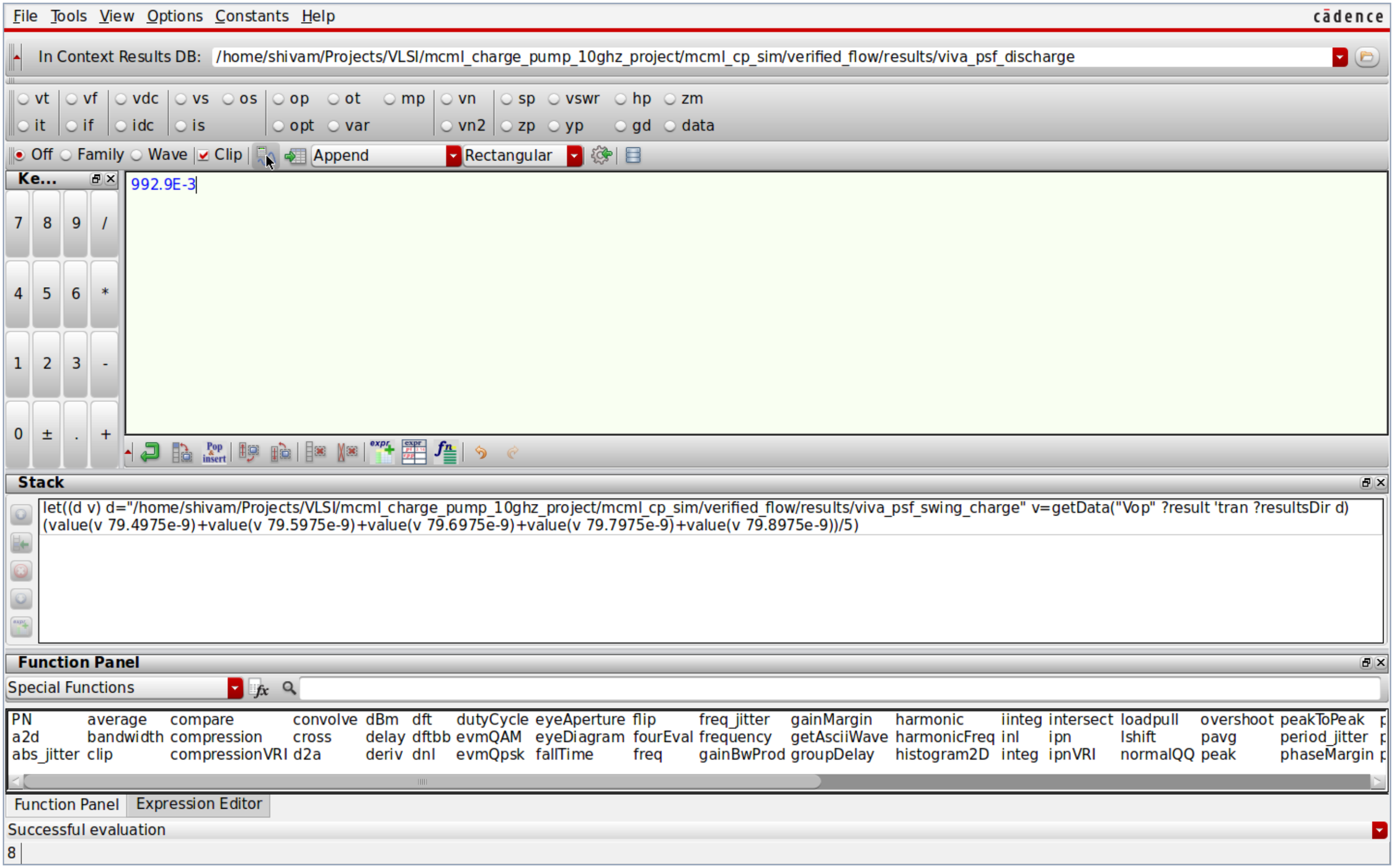
-average(I(V_VDD)) x 1.0 V from 1-10 ns = 22.528 uW; control sources are excluded.

Calculator: discharging VDD power



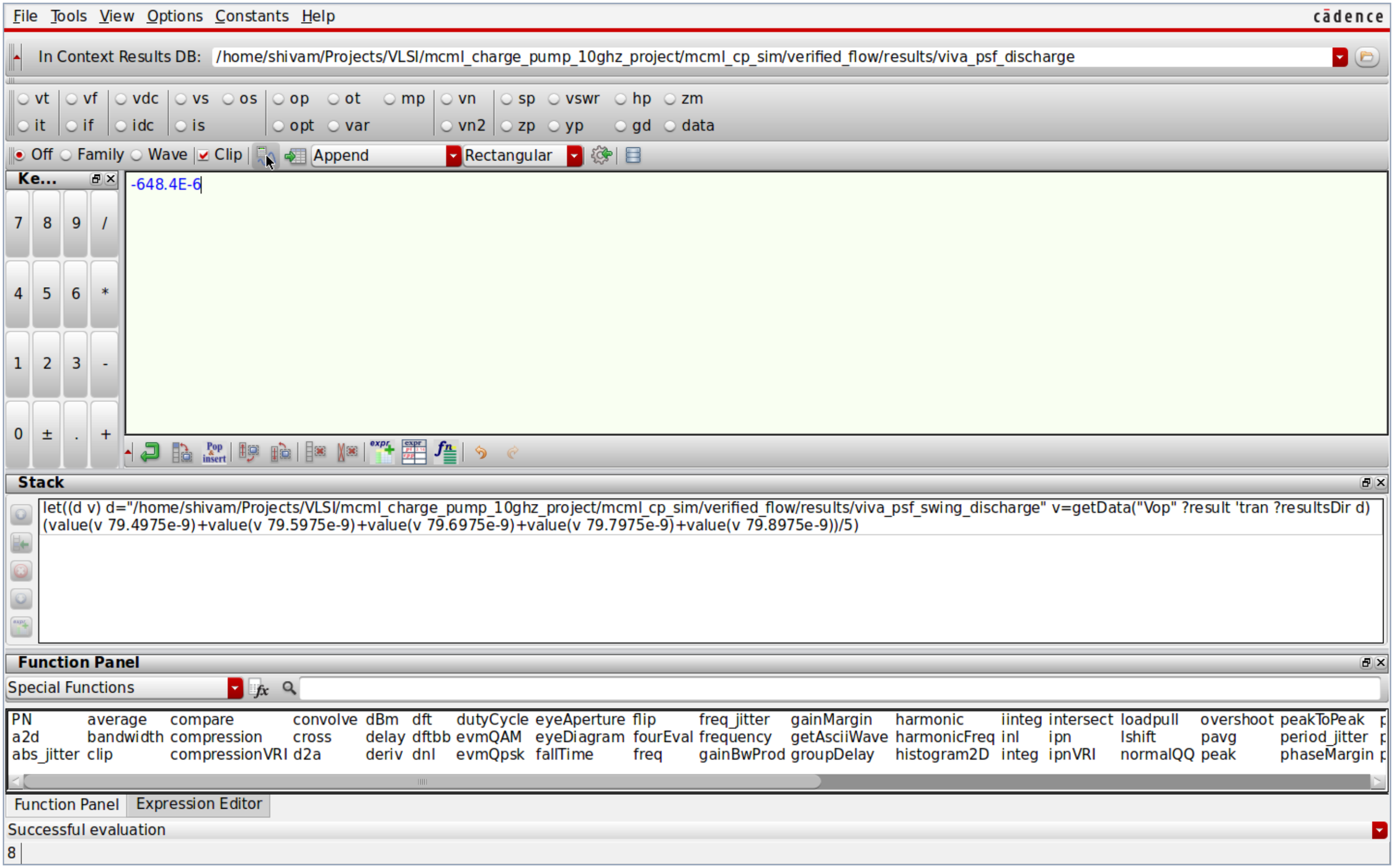
-average(I(V_VDD)) x 1.0 V from 1-10 ns = 16.097 uW; control sources are excluded.

Calculator: settled charging high level



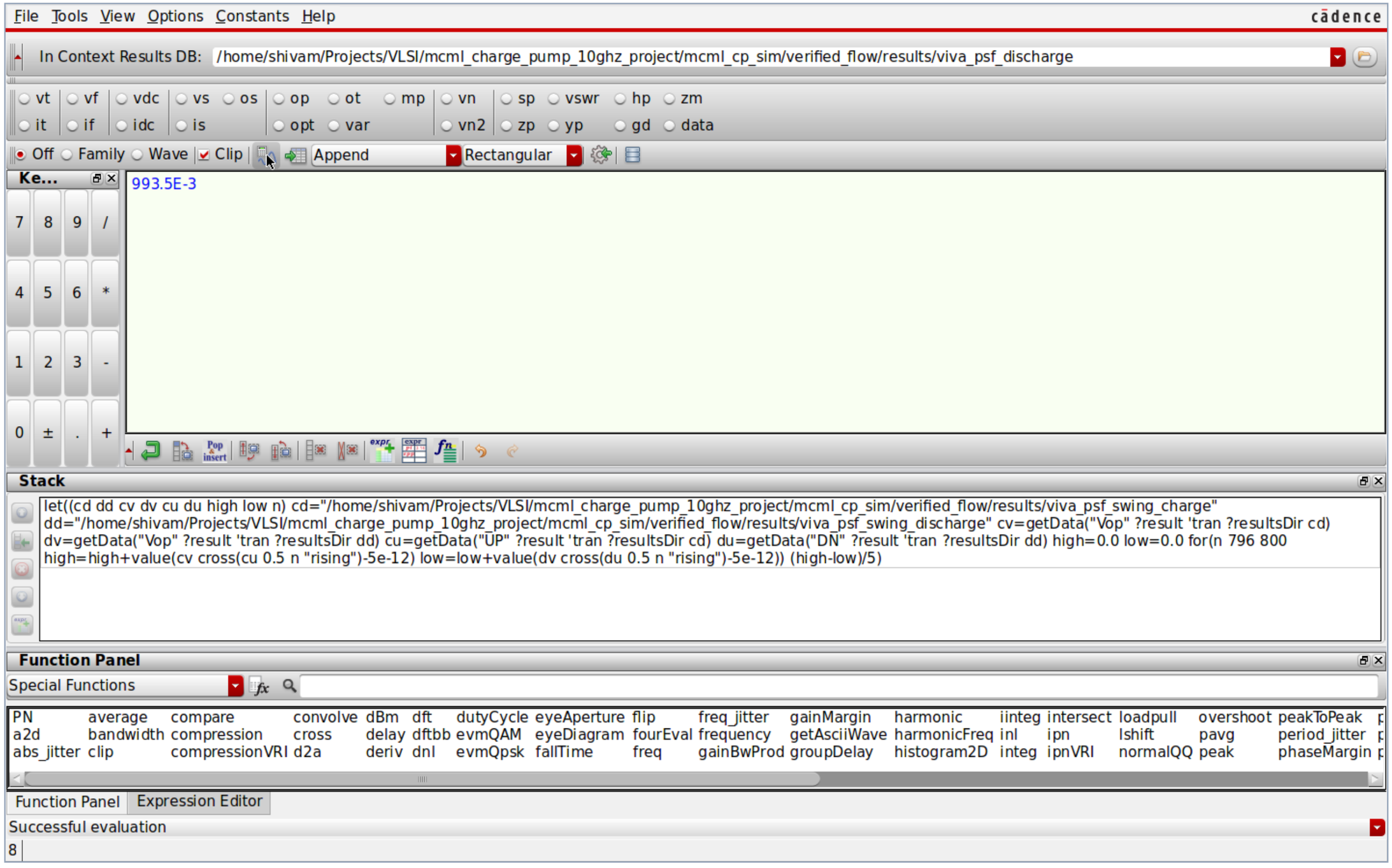
Five final quiet-time samples at 80 ns average 0.992875201 V.

Calculator: settled discharging low level



Five final quiet-time samples at 80 ns average -0.000648399 V.

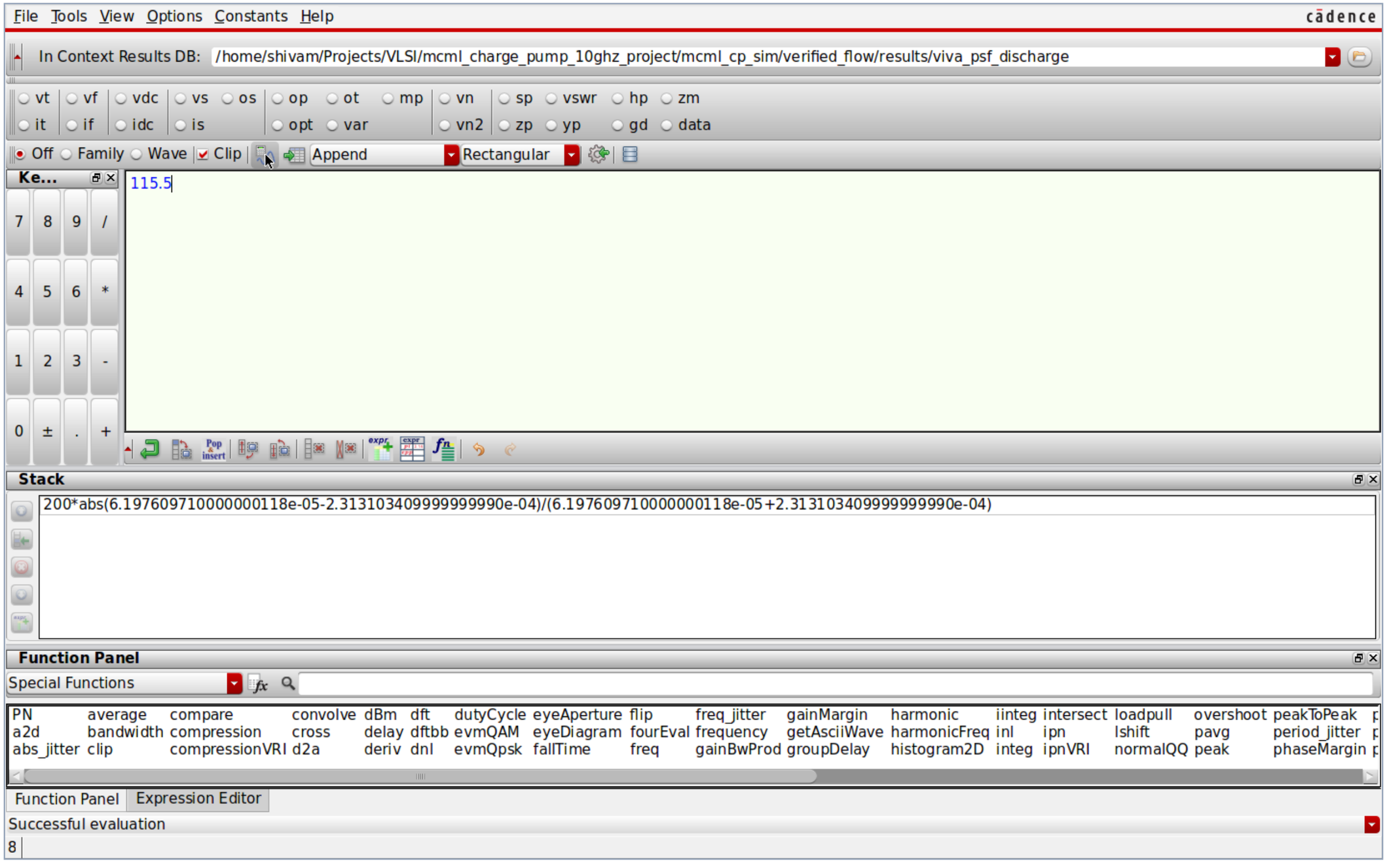
Calculator: settled output voltage swing



High minus low = 0.993523600 V (993.524 mV).

This is the transient output swing, which is distinct from DC current-compliance range.

Calculator: DC charge/discharge current mismatch



Calculator arithmetic uses measured DC-sweep currents: 61.976 and 231.310 uA at Vop=0.5 V.

Mismatch = 200*abs(Icharge-Idischarge)/(Icharge+Idischarge) = 115.47%.

The 10% criterion fails at its reference voltage; no valid current-compliance interval contains 0.5 V.